

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Experiments

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1510

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

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COURSE CODE: CSA1510

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Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Aim:

To create a simple cloud-based Library Book Reservation System for the SIMATS Library using Zoho Creator as a Cloud Service Provider and demonstrate the Software as a Service (SaaS) model by managing book details and reservation status online.

Procedure:

1. Open Zoho Creator and sign in to the account.
2. Create/open the Library Management System application.
3. Create a Book Details form for storing library information.
4. Add the required fields:
 - ☐ Title of the Book
 - ☐ Author
 - ☐ Publication
 - ☐ Year
 - ☐ Edition
 - ☐ No. of Copies
 - ☐ Rack No.
 - ☐ Status
5. Set the Status field as a dropdown with options such as Available, Taken,

and Returned.

6. Enter sample book records with their corresponding details.
7. Create a Book Report/List View to display all the stored records in tabular form.
8. Test the application by adding, viewing, and updating book details and reservation status.
9. Verify that the application can be accessed through the cloud using a web browser.

Output:

Library Management System

Dashboard

Books

Add Book

Reservations

Students

Reports

Settings

BOOKS

+ Add Book

Search books

Book ID	Title of the Book	Author	Publication	Year	Edition	No. of Copies	Rack No.	Status
BK001	Database Management Systems	R. Elmasri	Pearson	2024	7th Edition	5	R-12	Available
BK002	Computer Networks	Andrew S. Tanenbaum	Pearson	2023	6th Edition	3	R-08	Taken
BK003	Operating System Concepts	Silberschatz	Wiley	2023	10th Edition	4	R-15	Available
BK004	Cloud Computing	Rajkumar Buyya	McGraw Hill	2022	2nd Edition	2	R-21	Returned
BK005	Artificial Intelligence	Stuart Russell	Pearson	2024	4th Edition	3	R-18	Available

Welcome, Admin

Aim:

To create a simple cloud-based Payroll Processing System using Zoho Creator as a Cloud Service Provider and demonstrate the Software as a Service (SaaS) model by managing employee salary and payroll details online.

Procedure:

1. Open Zoho Creator and sign in to the account.
2. Create/open the Payroll Processing System application.
3. Create an Employee Payroll form to store employee details.
4. Add the required fields:
 - ☐ Name of the Employee
 - ☐ Experience in Present Organization
 - ☐ Overall Experience
 - ☐ Basic Pay
 - ☐ HRA
 - ☐ DA
 - ☐ TA
 - ☐ Gross Pay
 - ☐ Net Pay
5. Enter the employee details and salary components into the form.
6. Configure the payroll calculations:
 - $\text{Gross Pay} = \text{Basic Pay} + \text{HRA} + \text{DA} + \text{TA}$
 - $\text{Net Pay} = \text{Gross Pay} - \text{Deductions}$
7. Create an Employee Payroll Report to display the entered records in tabular format.
8. Test the application by adding multiple employee records and verifying the calculated salary values.
9. Access the application through a web browser to demonstrate cloud-based SaaS functionality.

Output:

Payroll Processing System

PAYROLL PROCESSING SYSTEM

Manage Employee Payroll Details

Welcome, Admin

Dashboard

Employees

Payroll

Generate Payslip

Reports

Settings

Total Employees

6

Total Gross Pay

₹ 924,500.00

Total Deductions

₹ 135,850.00

Total Net Pay

₹ 788,650.00

EMPLOYEE PAYROLL DETAILS

+ Add Employee

Import

Search employees...

Emp ID	Name of the Employee	Experience in Present Org (Years)	Overall Experience (Years)	Basic Pay (₹)	HRA (₹)	DA (₹)	TA (₹)	Gross Pay (₹)	Net Pay (₹)	Actions
EMP001	Ramesh Kumar	3.4	5.8	30,000.00	12,000.00	9,000.00	3,000.00	54,000.00	46,350.00	
EMP002	Priya Sharma	2.1	4.6	28,000.00	11,200.00	8,400.00	2,800.00	50,400.00	43,302.00	
EMP003	Arun Verma	4.7	7.3	35,000.00	14,000.00	10,500.00	3,500.00	63,000.00	53,805.00	
EMP004	Sneha Patel	1.5	3.2	25,000.00	10,000.00	7,500.00	2,500.00	45,000.00	38,610.00	
EMP005	Vikram Singh	5.2	9.1	40,000.00	16,000.00	12,000.00	4,000.00	72,000.00	61,560.00	
EMP006	Neha Gupta	2.8	5.0	27,000.00	10,800.00	8,100.00	2,700.00	48,600.00	41,023.00	

Result:

The Payroll Processing System was successfully created using Zoho Creator as a cloud-based SaaS application. The system stores employee experience and salary details such as basic pay, HRA, DA, TA, gross pay, and net pay. The payroll information can be managed, calculated, and accessed online through the cloud.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Aim:

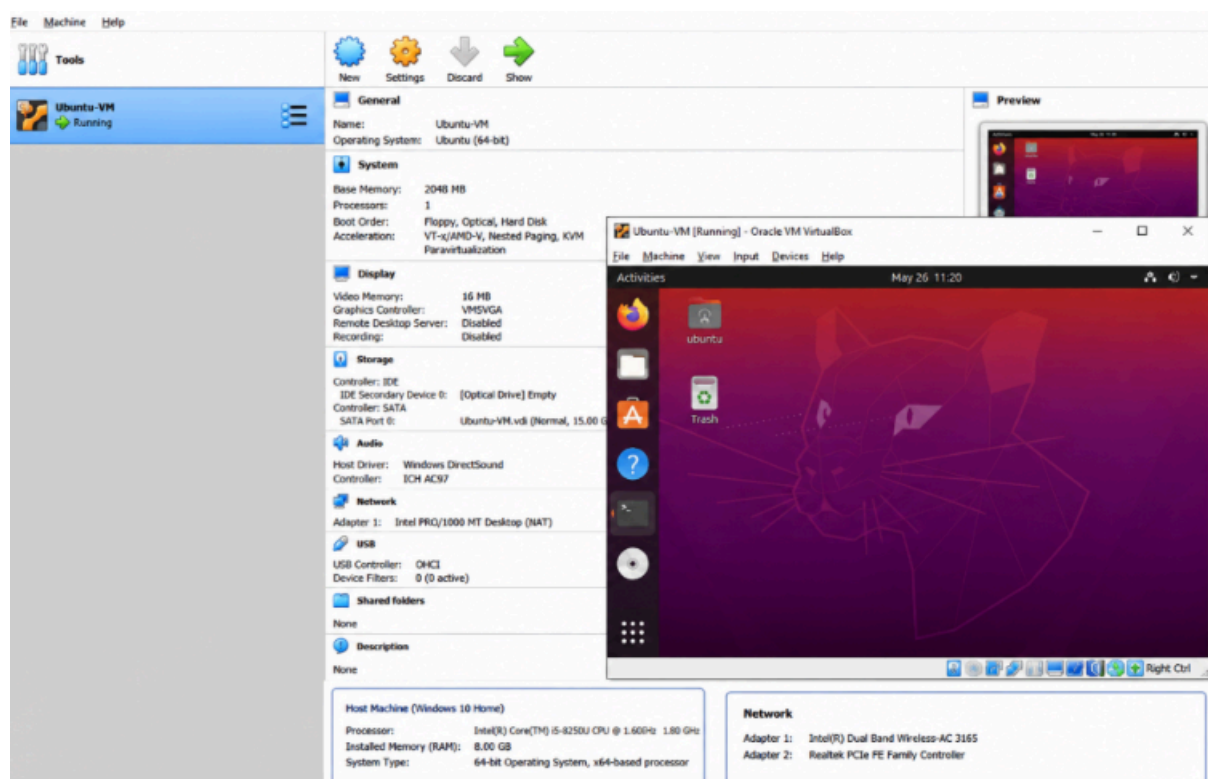
To demonstrate virtualization by installing a Type-2 Hypervisor and creating a Virtual Machine with 1 CPU, 2 GB RAM, and 15 GB storage using virtualization software such as Oracle VM VirtualBox.

Procedure:

1. Install Oracle VM VirtualBox, a Type-2 Hypervisor, on the host operating system.
2. Open VirtualBox Manager and click New to create a virtual machine.
3. Enter the VM name as Ubuntu-VM and select Linux/Ubuntu (64-bit) as the guest operating system.
4. Allocate 2 GB (2048 MB) of RAM to the virtual machine.

5. Set the processor configuration to 1 CPU.
6. Create a new virtual hard disk with a storage capacity of 15 GB.
7. Select the appropriate virtual disk type and attach the Ubuntu ISO image.
8. Start the virtual machine and install the Ubuntu operating system.
9. After installation, verify the VM configuration in VirtualBox Manager.
10. Run the virtual machine and confirm that the guest operating system works independently of the host operating system.

Output:



Result:

The Type-2 virtualization was successfully demonstrated using Oracle VM VirtualBox. A virtual machine named Ubuntu-VM was created with 1 CPU, 2 GB RAM, and 15 GB storage on the host operating system. The Ubuntu guest operating system was successfully installed and executed, demonstrating the use

of virtualization.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Aim:

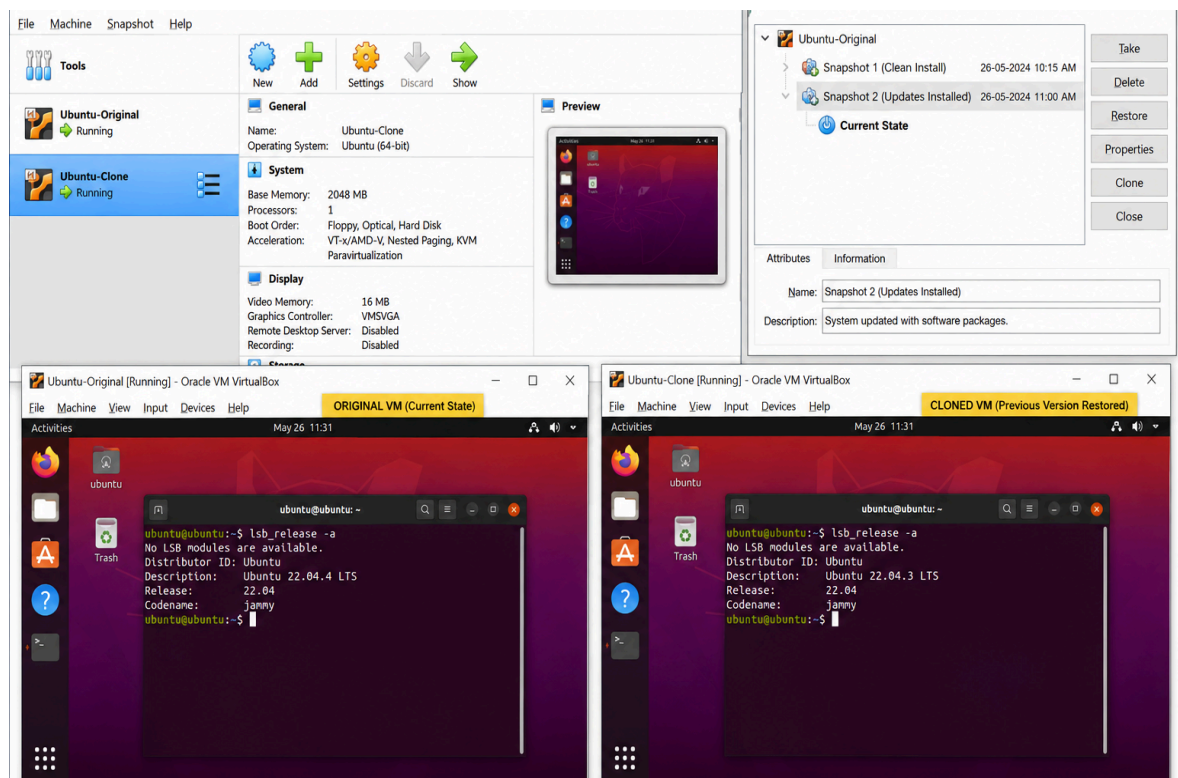
To demonstrate virtualization using a Type-2 Hypervisor by creating a Virtual Machine, cloning the VM, and testing the cloned/previous version by loading it successfully using Oracle VM VirtualBox.

Procedure:

1. Install and open Oracle VM VirtualBox on the host operating system.
2. Create a Virtual Machine named Ubuntu-Original and install the Ubuntu operating system.
3. Configure the VM with the required hardware resources and verify that it is working properly.
4. Take a Snapshot of the working VM to save its current state.
5. Right-click the Ubuntu-Original VM and select Clone.
6. Enter the name Ubuntu-Clone and select Full Clone.
7. Complete the cloning process and verify that the cloned VM appears in VirtualBox Manager.
8. Start the Ubuntu-Clone VM and check whether the operating system and files are loaded correctly.

9. Make a small change in the original VM and create another snapshot if required.
10. Use the Restore option in VirtualBox to load the previously saved version/snapshot.
11. Verify that the VM returns to the earlier saved state and that the cloned VM can also be started independently.

Output:



Result:

The VM cloning and previous-version testing was successfully demonstrated using Oracle VM VirtualBox. A working Ubuntu virtual machine was cloned successfully, and the cloned VM was loaded and tested independently. The previous VM state was also restored using the snapshot feature, demonstrating

VM cloning, snapshot management, and virtualization using a Type-2 Hypervisor.